

# SmartForge ML Assignment - Reflection Report

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Course: AI Fellowship - Week 9

Date: July 8, 2026

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## 1. Part 0: Neural Network Foundations

### Why ReLU is better than Sigmoid

Sigmoid squashes the output values between 0 and 1. The derivative of sigmoid is very small, maxing out at 0.25. When we build deep networks and backpropagate, we have to multiply these small derivatives together layer by layer. This makes the gradients in the early layers get extremely close to 0, which is the vanishing gradient problem. The early layers basically stop learning. ReLU solves this because for any positive input, its derivative is always 1.0. This means gradients can flow backward through the layers without shrinking, making the model train much faster.

### Cross-Entropy Loss vs. MSE Loss for Classification

MSE is designed for regression tasks where we predict continuous numbers. If we use it for classification with a softmax output, the gradient depends on the derivative of the softmax function. If the model makes a confident but wrong prediction, the softmax derivative becomes close to 0. This causes gradient saturation, meaning the model learns very slowly even when it makes a huge mistake. Cross-Entropy Loss works better because its math uses logarithms to penalize errors. The gradient is proportional to the actual error, so if the model makes a bad mistake, it gets a strong gradient update to fix itself quickly.

### Comparing SGD, Momentum, and Adam

Plain SGD is slow and bounces around a lot in noisy loss landscapes because it updates weights based only on the current batch. SGD with Momentum is like a rolling ball - it carries over some velocity from past steps, which helps it smooth out these oscillations and push past local minima. Adam is usually the best choice because it adapts the learning rate for each weight individually. It does this by keeping track of both the average gradient direction (momentum) and the squared gradient magnitudes (to scale the step size).

In terms of memory, SGD is the most lightweight since it does not store any extra velocity buffers. Momentum needs one extra copy of the model parameters to store velocity. Adam is the most memory-heavy because it needs two extra parameter-sized buffers to track the first and second moments of the gradients.

## 2. Part A: CNN Defect Classifier

### Why we use CNNs instead of MLPs for images

MLPs (standard dense networks) require us to flatten images into 1D vectors, which destroys all spatial relationships between pixels. Also, connecting a 200x200 image to a hidden layer of 128 nodes creates over 5 million parameters in just one layer, which quickly leads to overfitting. CNNs solve this by using small shared filters (like 3x3 kernels) that slide across the image. This captures local patterns like edges and textures, keeps the parameter count small, and makes the model translation-invariant (meaning it can recognize a defect no matter where it is on the steel sheet).

### Class Confusion: Crazing vs. Patches

During testing, the model often confuses Crazing (Cr) and Patches (Pa). This happens because both defects look like general surface textures rather than having sharp, clear shapes (like scratches or inclusions). Crazing is a

network of very fine cracks, while patches are regional color changes. Under different lighting angles or at lower resolutions, the fine lines of crazing can easily look like the shading variations of patches, making them hard for the model to separate.

### 3. Part B: Model Hardening

#### Why Data Augmentation is important

Data augmentation (like random flips, rotations, and crops) is the most effective way to improve generalization. In a real factory, steel sheets pass along conveyor belts at different angles, speeds, and lighting. By applying these transformations randomly during training, we force the model to learn the actual defect patterns rather than memorizing the exact positions of the training images. We only apply these to the training set, keeping the validation and test sets clean to ensure our evaluation is realistic.

#### BatchNorm and Dropout regularizers

Batch Normalization (BatchNorm2d) normalizes the outputs of convolutional layers so they have a mean of 0 and variance of 1. This keeps the inputs to deeper layers stable during training, allowing us to use higher learning rates and speed up convergence. Dropout randomly turns off a set percentage of neurons (like 40%) at each training step. This prevents the network from relying too much on specific neurons, forcing it to find multiple paths to learn the features. During evaluation (`model.eval()`), dropout is turned off so predictions are stable and consistent.

### 4. Part C: Hyperparameter Tuning

#### Grid Search findings: Learning Rate vs. Batch Size

Our grid search showed that the learning rate has a much bigger impact on validation accuracy than the batch size. If the learning rate is too high (like 0.01), the training loss bounces around or explodes because the steps are too large. If it is too low (like 0.0001), the model takes too long to learn. Batch size is less sensitive; smaller batches add some helpful gradient noise that can act as a regularizer, while larger batches are faster to compute but require more GPU memory.

#### Learning Rate Scheduler (StepLR)

Starting with a larger learning rate is useful early in training to explore the loss space. However, near the end, a high learning rate causes the weights to oscillate around the minimum. A scheduler like StepLR solves this by halving the learning rate every 5 epochs. This lets the model take smaller, more precise steps to settle directly into the best local minimum.

#### Optuna Bayesian Optimization

Manual grid search is slow because it checks hyperparameter combinations randomly or blindly. Optuna uses Bayesian Optimization to make smart guesses. It builds a probability model based on past trials to predict which values (like learning rate and batch size) are most likely to improve validation accuracy, finding the best settings with far fewer trials.